

Abdul Khurram

+1 (236) 865-4820 | rafay@abdulkhurram.com | linkedin.com/in/abdulkhurram | github.com/arkb75

EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo, Top 5% Internationally

2021

WORK EXPERIENCE

Shopify | *Codex, MCP, Shell, Nix*

Sept. 2026 – Present

Software Engineer Intern

Toronto, ON, Canada

- Built a **Codex integration** for internal AI tooling, now used across **several teams**, with 9 reusable skills and session hooks.
- Designed an **MCP bridge** to an internal coding agent; tested session discovery and revised the design after an API experiment ruled out full-session scans.

Contello.ai | *TypeScript/Node.js, AWS, Playwright, OpenAI*

Jan. – Dec. 2025

Software Engineer Intern

Toronto, ON, Canada

- Integrated **OpenAI** into production **TypeScript/Node.js** on **AWS**; built **agentic content-generation workflows** serving **thousands/day** with **observability** via **CloudWatch**.
- Enabled **AI-assisted content creation** by ingesting brand docs via an **NLP** pipeline, caching **embeddings**, and auto-injecting context while enforcing **responsible AI guardrails**.
- Strengthened **AI reliability** to **99.9%** via **OpenAI Responses API** migration, **cost/latency tracking**, **retry/backoff**, and **Playwright E2E** gates in **CI/CD**.

RBC Royal Bank | *Apache Spark, Python, Airflow, SQL, Linux*

Jan. – Aug. 2026

Software Developer Co-op

Vancouver, BC, Canada

- Engineered **Python/Airflow** pipelines processing **50M+ daily transactions** from **Delta Lake**, enabling downstream **ML/AI teams** with analysis-ready data on **petabyte-scale** financial systems.
- Optimized **Spark** jobs to cut **test runtime 40%**; rolled back the change after production instability.
- Worked on **multi-region** pipelines with **observability**, **fault tolerance**, and automated **failover**; instrumenting **cost/performance metrics** to inform infra decisions.

RESEARCH EXPERIENCE

UBC Emerging Media Lab | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

Software Developer & Team Lead

Vancouver, BC, Canada

- Led a **human-AI collaboration** project building a VR clinical simulator for nurse practitioners in Unreal/C++.
- Built **multithreaded C++/Python supervision** for AI patient monitoring and protocol enforcement; optimized inference to cut **latency 40%** and **costs 90%**.

PROJECTS

SoloPilot | *Python, TypeScript/Next.js, AWS, LangChain*

Personal Project

- Prototyping an **AI-powered** freelancer workspace with **agentic workflows** automating intake, scope analysis, proposal generation, and delivery.
- Designed an autonomous **evaluator→revisor agent loop (LangChain/Python)** with rubric-based **benchmarking**, improving quality through measured **AI performance** analysis.
- Shipped an email intake console rendering versioned proposal PDFs via a **Vision Model**, demonstrating practical **AI integration** into day-to-day workflows.

Backer | *TypeScript/Next.js, AWS DynamoDB, NextAuth*

Hack The Coast 2026

- Architected an **AI-powered** startup discovery platform (**TypeScript/Next.js, DynamoDB**) serving personalized investor feeds at scale.
- Built **ML training infrastructure**: supervised pipeline ingesting signals from **DynamoDB event streams** to train an **XGBoost** ranker.

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Python, Java, SQL, C/C++, Haskell (coursework), HTML/CSS

AI & Platforms: OpenAI API, LangChain, NLP/embeddings; Next.js, Node.js, Playwright; AWS (Lambda, DynamoDB, S3, SQS, CloudWatch); Git, Docker, CI/CD